

The assignment brief is subject to change. Please check the NILE site for updates regularly.



Assessment Brief

Module title:	Advanced AI and Applications	Module code:	CSY3060
Assessment code/title:	PJ2	Assessment weighting/word-limit:	50%/ 2000 words
Submission date:	27/05/2026	Feedback date:	24/06/2026
Module Leader:	Esther Jin	Resit date:	13/07/2026

Assessment Task:

Title (Question): Individual Course Project

Setting the scene:

This assessment focuses on developing a comprehensive understanding of advanced AI through:

1. An expanded literature review on the selected AI topic from the first project (PJ1).
2. An experimental project incorporating an *algorithmic improvement or variation* on the baseline method reproduced in PJ1.

This assignment requires you to:

- Review at least three additional relevant studies, advancements, recent developments or applications in the field.
- Design and propose an algorithmic improvement or variation on the baseline method. Clearly describe the proposed methodology.
- Implement the proposed method using PyTorch or TensorFlow in a systematic and reproducible manner.
- Analyse the performance of the designed improvement or variation. Compare the results against the baseline method.

The Task:

You must submit the following components:

1. A written report of approximately 2000 words (excluding references and appendix).
2. A working implementation (source code, .ipynb or .py file).
3. A demo video presentation (5-10 mins) that showcases your work. Use screen recordings and voice-over commentary.

Your written report should be structured to include at least the following sections:

0. Abstract
1. Introduction
2. Literature Review
3. Methodology
4. Experiments and Results
 - 4.1 Dataset/Environment
 - 4.2 Evaluation metrics
 - 4.3 Experimental results
5. Discussion and Conclusions

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6. References
7. Appendix (optional)

Report Requirements and Guideline

You are required to submit an individual report of approximately 2,000 words (excluding references and appendix). Your report should be structured under the following sections:

0. Abstract

- The topic and the specific problem being addressed (1-2 sentences)
- What this project aims to investigate (1 sentence)
- The motivation and core idea of the proposed method (2-3 sentences)
- Experimental evaluation including the dataset and environment, statistical results, and a comparison with the baseline method (2-3 sentences)

1. Introduction

- Introduce the topic — why it is important in AI research or applications.
- Introduce the reviewed literature briefly.
- Explain the motivation behind the proposed method — why the method can potentially improve the baseline method.
- Provide a brief description of the methodology.
- Experimental settings and performance — dataset/environment, results and comparisons.

2. Literature Review

- Summarise existing approaches and methods in the chosen area.
- Compare at least three research papers (or equivalent reputable sources).
- Clearly explain the models or algorithms in plain, conceptual terms, using concise language.
- Critically analyse the effectiveness and limitations of the reviewed approaches including their ethical and societal implications.
- Use figures or tables if helpful to illustrate comparisons.

3. Methodology

- Clearly describe the proposed approach in detail, using conceptual language, and support the explanation with equations where necessary.
- Specify the *exact modification*, design rationale, overall algorithm, model architecture, or learning strategy
- Include *diagrams or figures* illustrating the proposed method or model architecture, workflow, or algorithm flow.

4. Experiments and Results

- Describe the datasets (training and testing) or environments used in the experiments.
- Specify evaluation metrics (accuracy, loss, reward, etc.)
- Visualise the performance and compare the results against the baseline method, e.g., using loss curves, tables (statistical results), reconstructed or generated samples, agent trajectories, etc.
- Analyse and interpret the results:
 - Interpret the observed effects of the designed improvement or variation.
 - Critically discuss strengths and weaknesses of the proposed method.

5. Discussion and Conclusions

- Summarise the project and the key findings from experiments.
- Summarise effectiveness and limitations/challenges of the proposed method.
- Reflect on the ethical and societal implications of the proposed method.
- Suggest potential improvements, extensions, or alternative approaches.

6. Reference

- Include all sources cited in the literature review and methodology sections.
- Using the Harvard referencing style.

7. Appendix (optional)

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- Detailed tables or figures that support results
- Additional diagrams or model visualisations

Video Demonstration:

- You are required to submit a video demo (URL) of your assignment, 5-10 mins.
- Ensure the video link is accessible to the marker (do not set it to private).

Please ensure that you keep a secure electronic backup copy of your assignment.

NOTE: Students who cannot demonstrate understanding of their work will not pass.

The module tutor may invite you for a viva-voce. Poor demo/viva could negatively influence other sections in the marking criteria and result in an overall fail grade.

Learning Outcomes aligned to this Assessment:

On successful completion of this assessment, you will be able to:

- LO-c. Critically appraise, to a satisfactory level, the various AI methods, considering their appropriateness, advantages and disadvantages in particular application.
- LO-d. Design, develop and evaluate a range of typical complex applications using artificial intelligence methods.
- LO-e. Implement an ethically sound solution to a complex problem.
- LO-f. Critically analyse techniques and resources for meeting the prioritised requirements of a complex task.

Academic Practice (referencing style, literature usage, AI Usage):

You must reference all sources appropriately using the Harvard referencing style. Failure to acknowledge sources may constitute academic misconduct. You are allowed to use research papers and technical reports especially in the related works section.

Assessment Guidance:

Reading List: You will find a link to your online reading list on NILE

Use of Generative AI (Artificial Intelligence) within this Assessment:

AI Categories:

Category 2: GenAI can be used in an assistive role

You may make use of GenAI in your assessment in an assistive role, but you must acknowledge this appropriately. Steps 2 and 3 outline the different types of GenAI use, what you need to acknowledge and how you should do this.

Note: See University Of Northampton Guidance on GenAI:

<https://libguides.northampton.ac.uk/artificialintelligence/guidance>

Academic Practice support

The Skills Hub is a central repository where you will find a range of support for your study and assessments: <https://skillshub.northampton.ac.uk/>

Feedback:

Feedback should be received within 20 working days

An announcement will be sent out via NILE to inform you of when feedback is available.

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Instruction on Anonymity for students [Further guidance is available online](#)

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CSY3060: Undergraduate Marking Rubric

Assessment Criteria	Weight	Evidence checklist	Learning Outcomes	Fail Work does not achieve requirements	Pass Work achieves requirements	Satisfactory Work of satisfactory quality	Good Work of worthy quality	Excellent Work of high quality
Abstract & Introduction	10%	Problem definition, context, motivation, rationale	c	No clear problem defined; lack context, motivation, or rationale	Problem identified but vague, poorly motivated, or weakly contextualised	Problem, context, and motivation adequately introduced with some rationale and contextual background	Clear and relevant problem; motivation well explained and situated in context	Precise problem definition; clear motivation; strong rationale linked to real-world or research impact
Literature Review	15%	Coverage of sources, comparison of approaches, quality of sources	c, f	No review; irrelevant or missing sources	Minimal review; weak or non-academic sources	Adequate review with some relevant academic sources	Good review; sources relevant and critically compared	Excellent review with strong critical insight and wide coverage of sources
Methodology	20%	Description of method, modification, diagrams or figures, feasibility, creativity	d, f	No clear methodology; lack of diagram/figure; missing or incorrect methodology details	Basic methodology variation, but vague or superficial; diagram/figure missing or unclear	Adequate methodology in mostly clear terms; minor creativity	Good methodology; appropriate diagram/figure used to support understanding	Excellent methodology; feasible; creative; clear presentation; excellent diagram/figure with insightful explanation
Implementation & Artifact	20%	Implementation of the proposed method, use of PyTorch/TensorFlow, functional artifact, clear code structure, readability & documentation	e	No implementation provided; code does not run or does not reflect the proposed method	Minimal or incomplete implementation; artifact does not function; neither PyTorch nor TensorFlow is used	Adequate implementation; artifact functions with some issues	Good quality implementation; functional artifact with clear documentation	High-quality implementation; artifact highly functional, and well-documented
Experiments & Results	15%	Experimental design, results presentation, performance comparison & discussion	d, e, f	No experiment design; results missing or unrelated	Minimal or weak experiment attempted; results presented with little clarity or interpretation	Adequate experimentation with basic results and some discussion	Well-designed experiment with clear purpose; results clearly presented; sensible interpretation	Insightful and well-motivated experimental design; results presented clearly and analysed critically
Ethics & Critical Evaluation	10%	Ethical issues, legal/environmental considerations, mitigations	c	No mention of ethics/legal/environmental issues	Minimal mention without relevance	Adequate coverage of some issues but limited depth	Good coverage of relevant issues with suggested mitigations	Excellent, insightful analysis of ethical/legal/environmental risks with strong mitigation strategies

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Presentation, Referencing & Demo Video	10%	Report organisation, clarity, academic writing, Harvard referencing, demo video	c, d, f	Report missing or incoherent; no referencing; demo video missing or video unrelated	Poorly structured; limited referencing; demo video unclear or poorly explained	Adequately structured; some referencing errors; demo video with limited explanation	Well-structured report; consistent referencing; clear demo video showing key results with appropriate explanation	Excellent structure and clarity; flawless referencing; high-quality demo video with clear presentation and insightful explanation
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